

AN INTERNSHIP REPORT
ON
Quality Assurance
At
Avoloft Technologies Pvt. Ltd.
Tinkune, Kathmandu

BY
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An Internship Report Submitted to
Faculty of Management, Tribhuvan University
in the partial fulfillment of the requirements for the degree of
Bachelor of Information Management

Kathmandu, Nepal

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STUDENT DECLARATION

This is to certify that I have completed the report entitled “**Internship Report on Quality Assurance at Avoloft Technologies Pvt Ltd**” under the guidance of “**Mr. Nabin Dhital**” in partial fulfillment of requirements for the degree of **Bachelor of Information Management (BIM)** at Faculty of Management, Tribhuvan University. This is my original work, and I have not submitted it earlier elsewhere.

Date: June 2025

Signature:

Nibesh Upadhyaya

LETTER OF APPROVAL

RECOMMENDATION LETTER

ACKNOWLEDGEMENT

The internship opportunity at Avoloft Technologies Pvt. Ltd. has been valuable experience for learning and professional development. This report is prepared as a partial fulfillment of the requirements for the Bachelor in Information Management (BIM) under the Faculty of Management, Tribhuvan University. I would like to express my sincere thanks to everyone who supported me throughout this journey.

First and foremost, I would like to express my deepest appreciation to my academic supervisor, Mr. Nabin Dhital, for his guidance and encouragement during the internship period. I am also thankful to my mentor at Avoloft, Mr. Laxman Magarati, for his continuous support, clear direction, and valuable insights.

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Finally, I would like to thank Tribhuvan University for introducing the internship program and giving students the opportunity to experience real-world work settings. I also extend my thanks to Shanker Dev Campus for the continuous support and for providing the platform to complete this internship successfully.

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ABSTRACT

This internship report reflects my three-month journey at Avoloft Technologies Pvt. Ltd., where I worked as a Quality Assurance (QA) intern. The experience served as a bridge between classroom learning and real-world application, giving me the chance to understand how software is tested and improved before reaching end users. Guided by a supportive team and mentor, I explored the QA process from the ground up and observed how quality is maintained throughout the SDLC.

During the internship, I engaged in a range of hands-on activities. I practiced manual testing, created and executed test cases, and documented bugs. I also explored API testing using Postman and dipped into automation testing using the Cypress framework. Along the way, I became familiar with important testing techniques like Boundary Value Analysis and Equivalence Partitioning, and I learned how version control systems like Git and platforms like GitHub.

Although I wasn't directly involved in live projects since my internship started due to company policies, the simulated tasks and structured guidance gave me real insight into how QA teams operate. This internship not only strengthened my technical skills but also helped me grow professionally learning the importance of teamwork, attention to detail, and continuous improvement. It's an experience that has solidified my interest in pursuing a career in software testing and quality assurance.

Keywords: Internship, Quality Assurance, Manual Testing, Automation Testing, API Testing, Cypress, Postman, Test Case Design, Bug Reporting, Version Control, Git, Software Development Lifecycle (SDLC).

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ABBREVIATION

API	Application Programming Interface
BVA	Boundary Value Analysis
CRUD	Create, Update, Read, Delete
CSS	Cascading Style Sheet
QA	Quality Assurance
QTP	Quick Test Professional
SDLC	Software Development Life Cycle
SQ	Software Quality
STLC	Software Testing Life Cycle
TU	Tribhuvan University

CHAPTER ONE: INTRODUCTION

1.1 Background

An internship serves as a crucial bridge between classroom learning and real-world application, offering participants the chance to collaborate with seasoned professionals. This hands-on experience empowers interns to translate theoretical concepts into practical solutions, gaining a deeper appreciation for the complexities of their chosen field. Under the mentorship of experts, interns engage in projects, tasks, and challenges that foster skill development and industry understanding.

During an internship, individuals work under the guidance and supervision of experienced professionals in their field of interest. The primary purpose of an internship is to provide hands-on learning opportunities, allowing interns to apply the theoretical knowledge they have acquired in their academic studies to real-world situations.

This report details my experience as a Quality Assurance Intern at Avoloft Technologies Pvt. Ltd. My internship is focused on learning and experiencing how software systems work and various testing techniques, rather than participating in specific projects. Due to company policies, new and inexperienced interns like me were not involved in confidential or sensitive projects to maintain privacy and security.

Additionally, I learned about the roles and responsibilities of different team members in a software development environment, such as Business Analysts, Developers, Architects, Project Managers, DevOps Engineers, and QA Engineers. This comprehensive exposure helped me understand the importance of collaboration and the workflow within SDLC. I explored test design techniques like Boundary Value Analysis, Equivalence Partitioning, and Use Case Testing, managed test data, and conducted frontend testing. My training also covered API testing and Automation testing.

This internship has been instrumental in bridging the gap between theoretical knowledge and practical application, enhancing my skills in software testing and preparing me for a future career in this field.

1.2 Objectives

The objective of the internship was to gain practical experience in QA, while completing assigned tasks and projects to enhance skills and knowledge in the field.

1.2.1 General Objectives

- i. To learn new skills and acquire practical knowledge.
- ii. To develop industry-specific expertise and gain hands-on experience.
- iii. To enhance communication, teamwork, and problem-solving skills.
- iv. To receive feedback and guidance to improve professional growth.
- v. To explore career options and gain insight into the chosen field.

1.2.2 Specific Objectives

- i. To follow automation standards, best practices, and version control in software development.
- ii. To gain practical experience in various software testing methodologies.
- iii. To develop a clear understanding of quality assurance principles and practices.
- iv. To master debugging and troubleshooting for problem-solving.
- v. To become proficient in writing test cases, executing them, and reporting bugs effectively.
- vi. To familiarize myself with different testing tools and techniques, including API testing and test data management.

1.3 Methodology

The methodology section of the report provides a detailed overview of the internship approach and process, showcasing the structure and activities undertaken. It allows readers to understand the internship experience, assess the strategies used, and evaluate the achieved objectives. It is essential for transparency and showcasing personal and professional growth.

1.3.1 Organization Selection

As I had a keen interest in QA and a strong desire to enhance my skills in the field, I actively sought out opportunities to expand my knowledge and improve my resume. I dedicated time to self-study, utilizing documentation and YouTube channels to deepen my understanding. Simultaneously, I conducted research on various companies and organizations that prioritized open-source technologies as their core domain.

I applied to numerous companies and eagerly awaited their responses. Some companies acknowledged my resume and contacted me for interviews. My best efforts and hard

work led me to convince them during the interview process along with a small task of testing a Login module.

I successfully cleared the interview and task.

1.3.2 Placement

Fortunately, my performance during the interview left a positive impression. Same day at the evening, I received a call from the HR department congratulating me on my selection and offering me the position as an intern at the company for 3 months from December 17, 2024 – March 16, 2025

1.3.3 Duration

During the orientation day, the company provided me with detailed information about my assigned time schedule and the duration of my internship period.

Table 1.1 Duration

Start Date	December 17, 2024
End Date	March 16, 2025
Duration	3 Months
Days of Work	Sunday to Friday
Working Hour	10:00 AM to 6:00 PM

1.3.4 Activities Performed

Due to company policies regarding privacy and security, I was not directly involved in the project. However, my training at Avoloft Technologies included several key areas designed to simulate real-world testing scenarios and equip me with practical knowledge. I got to work on lot of activities which are listed below:

- **Manual Testing:** Engaged in manual testing of sample applications. This involved creating detailed test plans and test cases, executing these test cases, and identifying and reporting bugs. I performed frontend testing on various web applications, practiced finding locators for UI-related tests, and documented defects comprehensively.
- **API Testing:** Learned to perform API testing using Postman by creating and executing various API requests along with test scripts. This experience helped me understand how to validate the functionality, reliability, and performance of APIs.

- **Test Case Writing and Bug Reporting:** Practiced writing comprehensive test plan and test cases for different testing scenarios and learned how to document and report bugs effectively using industry-standard tools.
- **Frontend Testing:** Conducted frontend testing on various web applications, identifying, and documenting defects.
- **Test Design Techniques:** Explored test design techniques like Boundary Value Analysis (BVA), Equivalence Partitioning, and Use Case Testing, managed test data, and conducted frontend testing.
- **Automation:** Executed test scripts and automated tests using JavaScript as a programming language. Executed the test scripts and generated the test results automatically by using automation tools.
- **Version Control:** Gained experience with version control using Git and working with platforms like GitHub.

These structured training modules and practice tasks provided me with a comprehensive understanding of the software development and testing lifecycle. While I was not directly involved in client projects, the practical exercises and training sessions were instrumental in building my skills and preparing me for future contributions to real- world projects.

1.3.5 Tools Used

Over the course of the internship, I utilized a range of tools, outlined as follows:

- **Programming Language: JavaScript, Cypress Framework**
JavaScript is a versatile scripting language widely used for web development. Cypress is a powerful end-to-end testing framework built on JavaScript, designed for reliable and fast frontend testing.
- **System Used: Microsoft Windows 10 64-bit.**
Windows 10 64-bit is a widely adopted operating system known for its stability and compatibility with diverse software tools. It provides a robust environment for software development and testing.
- **API Platform: Postman**
Postman is a popular platform used for designing, testing, and managing APIs efficiently. It simplifies API development by offering a user-friendly interface for sending requests and analyzing responses.

- **Version Control: GitHub**

GitHub is a cloud-based platform for version control using Git, enabling collaboration and code management. It allows tracking changes, branching, and merging software projects.

- **Performance Testing: JMeter**

JMeter is an open-source tool designed to test the performance and scalability of applications under various load conditions. It helps identify bottlenecks by simulating heavy user traffic.

- **IDE for testing: IntelliJ IDEA 2024.1**

IntelliJ IDEA is a powerful integrated development environment (IDE) known for its intelligent coding assistance. The 2024.1 version offers enhanced features to streamline software testing and development workflows.

CHAPTER TWO: INTRODUCTION OF THE INDUSTRY

2.1 Overview

After the Information Revolution, there was the emergence of new Industry as IT Industry. IT industry simply can be said as industry that uses digital and modern technologies with the use of computers, devices, and other electronic devices. The IT industry is the technology-driven sector focused on creating, utilizing, and maintaining computer networks, software, and systems for processing and delivering data. It involves the integration of computer and telecommunications technology to bring about transformative change. As the fastest-growing economic activity globally, the IT sector holds a prominent position among industries.

Software is the backbone of any technology, as it enables the functionality and operation of systems. A technology without software is nothing but a trash. Computer software comprises data and computer instructions, including programs, libraries, and non-executable data like online help and digital media. It is an essential component of computer systems, working in conjunction with physical hardware.

Computer hardware and software are interdependent, relying on each other for effective operation. Hardware provides physical infrastructure, while software enables the execution of tasks and processing of data. The IT industry recognizes the critical relationship between hardware and software, emphasizing their interconnected nature. In summary, the IT industry encompasses the development, use, and management of computer networks, software, and systems. It is driven by the integration of computer and telecommunications technology and relies on the interdependency between hardware and software to deliver transformative solutions and drive economic growth.

2.2 History of IT

The domain of Information Technology (IT) has undergone a profound and far-reaching transformation on the global scale, reshaping the very fabric of modern societies and economies. Within this expansive evolution, the role of programming has emerged as a fundamental force propelling the advancement of the global IT industry. From the foundational development of programming languages to the intricate specialization across a multitude of domains, programmers worldwide have been instrumental in driving innovation, efficiency, and transformative solutions across industries. This growing expertise in programming has positioned the global IT sector to navigate the

intricacies of an ever-evolving technological landscape, fostering continuous growth and innovation in a world increasingly defined by digital frontiers.

The trajectory of IT history reflects the worldwide journey of integrating technology to unleash its potential for comprehensive development. Beginning with the early embrace of computers and culminating in the present era of pervasive digitalization, the global IT sector has been a central driver of progress. With strategic investments in infrastructure, progressive governmental policies, and a keen focus on enhancing skill sets, the global IT industry stands at the forefront of shaping the future, enabling societies to harness the full spectrum of technological possibilities and usher in a new era of innovation and prosperity.

2.3 Current scenario of IT in Nepal

IT is one of the rapidly growing industries in Nepal, which could further show very promising potential for growth even in the near future. The industry has grown successively and has seen many developments and opportunities in recent times. It contributes to the economic development and digital transformation of the country. 7 This growth has been manifested in the increasing rise of different technologies within various sectors, one of which is e-commerce. With innovation within this sector in development, cities like Kathmandu and Pokhara have emerged as tech hubs. Software development companies belonging to Nepal have made their contributions in this regard, offering solutions that range from web and mobile applications to enterprise systems. The IT industry is also getting government support, which is helping in the transformation of Nepal into a digitally empowered society. Besides, there are some IT companies which become successful to establish their own renowned name in the Nepal IT market. Some of the known IT companies of Nepal are:

- Leapfrog Technology, Inc.
- EB Pearls Pvt. Ltd.
- Logpoints Nepal
- F1 Soft International Pvt. Ltd.
- Fusemachines Nepal Pvt. Ltd.

2.4 Challenges and opportunities of IT in Nepal

There are so many challenges and opportunity that organizations have been facing since long. Some opportunity and challenges that have been seen till today are explained below:

2.4.1 Challenges

The Nepalese IT industry faces the following challenges to exist in the market, although it shows higher market growth in Nepal. The Challenges are:

- Budgeting for IT is not sufficient.
- Unfair market competition.
- There is no effective cyber law.
- Unpredictable government standards.

2.4.2 Opportunities

Nepal is a developing country provided that in IT sector Nepal is rapidly growing day by day in order to meet world technology. In Nepal there are several opportunities in the IT sectors. They are:

- High quality IT education market.
- High number of students pursuing Computer Science and IT education.
- Huge market, less innovation.
- Increasing number of working people.
- Low development costs.
- Investment from government in the field of research and development of IT.

2.5 Scope of Information Technology

Career in information technology is considered one of the most high-paying jobs and is full of opportunities. As a name of an undergraduate degree program, it refers to the preparation of students to meet the computer technology needs of business, government, healthcare, schools, and other kinds of organizations. The following are the list of common careers for IT graduates.

- UI/UX Designer
- IT Project Manager
- Cybersecurity Analyst
- Software Developer

- DevOps Engineer
- Network Engineer
- Artificial Intelligence (AI) / (ML) Engineer
- Database Administrator (DBA)
- Quality Assurance Engineer

2.6 Introduction to QA Industry

Quality Assurance (QA) in the Information Technology (IT) industry is a critical discipline focused on ensuring that software and systems meet specified requirements and function as intended. QA is the production department responsible for identifying defects and preventing failures. The primary goal is to deliver the highest quality to clients while ensuring that a product or service meets their expectations. QA involves a systematic process of evaluating and improving software quality through various testing methodologies and best practices. These processes aim to detect and correct defects before software deployment, ensuring reliability, efficiency, and user satisfaction.

QA encompasses a broad spectrum of activities, including manual testing, automated testing, performance testing, and security testing. Manual testing involves testers manually executing test cases without using automation tools, whereas automated testing leverages software tools to execute pre-scripted tests efficiently. Performance testing assesses the speed, responsiveness, and stability of a system under varying conditions, and security testing identifies vulnerabilities to ensure system security against potential threats.

In the IT industry, QA is an integral part of the software development lifecycle (SDLC). It begins in the early stages of development, often during planning and design, to ensure quality is embedded from the outset. Agile and DevOps methodologies emphasize continuous testing and integration, making QA an ongoing process rather than a final step. With the increasing role of software in multiple industries and the rise of agile development approaches, the QA industry has experienced significant growth in recent years.

To meet global standards, robust QA processes are necessary to ensure software reliability and functionality. Several IT firms in Nepal employ specialized QA professionals who work closely with software developers and project managers. Their primary objective is to ensure that software products meet high-quality standards, align

with user specifications, and adhere to predefined timelines and budgets. QA has become an essential and highly sought-after skill among Nepalese software professionals. QA practices in Nepal involve a structured approach to software testing and validation, ensuring the identification and resolution of issues before deployment. This guarantees that applications are efficient, reliable, and user-friendly. Various testing methods used in Nepal include:

- **Manual Testing:** Testers manually execute test cases to ensure expected software behavior from a user's perspective.
- **Automated Testing:** Automation tools such as Selenium, Cypress, Playwright and JUnit facilitate efficient execution of repetitive test cases, enhancing testing speed and accuracy.
- **Performance Testing:** Evaluates application performance under different load conditions to ensure stability and responsiveness.
- **Security Testing:** Identifies vulnerabilities to protect software applications against cybersecurity threats.

In Nepal, QA is integrated early into the software development lifecycle (SDLC), ensuring consistent quality throughout the development process. The increasing adoption of QA methodologies in Nepal's IT sector underscores its importance in delivering high-quality, secure, and efficient software solutions. Educational institutions and training centers in Nepal offer specialized QA training programs to equip professionals with the necessary skills to excel in this field.

CHAPTER THREE: INTRODUCTION OF THE ORGANIZATION

3.1 About Avoloft Technologies

Avoloft Technologies Pvt. Ltd. is a Nepal-based technology company established in 2023 by a team of five passionate individuals who came together with a shared vision: to transform creative ideas into meaningful digital solutions. Although relatively new to the industry, Avoloft has quickly gained attention for its innovative mindset, collaborative spirit, and strong commitment to understanding and meeting client needs. What sets Avoloft apart is its belief in building lasting partnerships through practical, reliable, and forward-thinking technology. The company operates with a people-first approach valuing quality, adaptability, and open communication. These values have helped Avoloft grow steadily, earning trust and recognition in a competitive digital landscape.

With a clear focus on digital empowerment and long-term value creation, Avoloft Technologies continues to position itself as a dependable and visionary technology partner for businesses looking to grow and innovate.

Table 3.1 Overview of Company Profile

Organization Name	Avoloft Technologies Pvt. Ltd.
Address	Tinkune, Kathmandu
Contact	+977 9842781897
Year of Establishment	2023
Email	avoloft.tech@gmail.com
Website	https://avoloft.com/

3.2 Mission

Avoloft Technologies Pvt. Ltd. is dedicated to empowering businesses of all sizes and industries with innovative digital solutions that drive growth, enhance online presence, and create lasting impressions. By transforming ideas into reality through cutting-edge technology and creative solutions, Avoloft aims to deliver top-notch services that meet the unique needs of each client.

3.3 Vision

Avoloft envisions shaping tomorrow's technology by empowering businesses with cutting-edge solutions, creative excellence, and a commitment to driving growth and success. Their goal is to innovate the future together with clients, turning bold ideas into impactful digital realities.

3.4 Values

- **Innovation:**
Lead to make services profitable, sustainable and relevant to market needs.
- **Proactiveness:**
Foster excellence through continuous improvement and a high level of accountability
- **People:**
Encourage a rich talent pool through trust, strong work ethic, and a conducive work environment
- **Work Beyond Boundaries:**
Collaborate and coordinate to share, learn and develop solutions and partnerships that are win-a-win with a shared vision.

3.5 Organization Structure

An organizational structure is a framework that describes how certain activities are to be carried out in order to accomplish an organization's objectives. These activities include rules, positions, and obligations and define the standard procedures and practices.

The following chart shows the organizational hierarchy of the organization where I completed my internship which is Avoloft Technologies Pvt. Ltd.

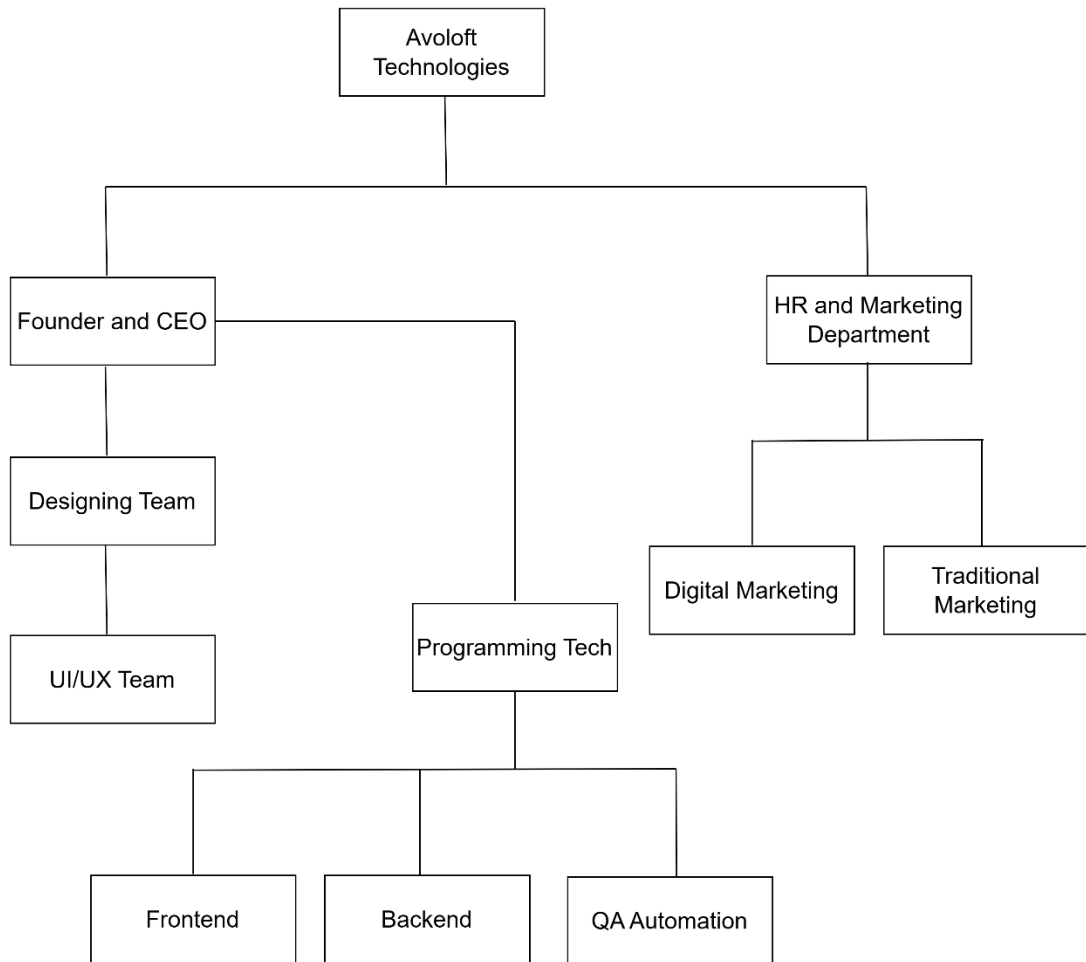


Figure 3.1 Organization Structure

3.6 Services Provided

The service provided by Avoloft Technologies Pvt. Ltd. are:

- **Web Development:** Crafting custom, responsive websites tailored to meet specific business objectives.
- **App Development:** Developing native and cross-platform mobile applications to enhance user engagement.
- **UI/UX Design:** Designing intuitive interfaces and seamless user experiences for improved interaction.
- **Digital Marketing:** Enhancing online presence through SEO, social media strategies, and targeted campaigns.
- **ERP Software:** Building customized enterprise resource planning systems to streamline business operations.

- **CMS & CRM Development:** Creating robust content and customer relationship management platforms.

3.7 Location

The primary office of Avoloft Technologies Pvt. Ltd. is situated at the following address:

Avoloft Technologies Pvt. Ltd.

Registered Office

Tinkune, Kathmandu, Nepal



Figure 3.2 Location of Avoloft Technologies Pvt. Ltd.

CHAPTER FOUR: ANALYSIS OF ACTIVITIES DONE

4.1 Analysis of Activities Performed

During my internship at Avoloft Technologies, I was not assigned to specific client projects due to the company's internship policy, which focuses on familiarizing interns with various aspects of software development and testing rather than involving them in ongoing projects. This policy ensures that interns build a strong foundation in essential skills before contributing to live projects

4.2 Roles and Responsibilities

During my internship period, I was under the supervision of the Founder and CEO of the company. Under his supervision, I was assigned various tasks focused on furnishing my knowledge on QA. I undertook various roles and responsibilities that allowed me to gain a comprehensive understanding of software testing and quality assurance practices. My primary focus was on learning and experiencing how software systems work, familiarizing myself with various testing techniques, and contributing to the quality assurance processes. Below are the key roles and responsibilities I handled during my internship:

- Gained a solid understanding of testing fundamentals including the different types of testing such as functional, non-functional, regression, sanity, integration, and unit testing.
- Performed manual testing activities, which involved writing test cases, executing them, and reporting bugs.
- Conducted frontend testing on various web applications, identifying and documenting defects
- Practiced finding locators and using them to write detailed test cases for user interface (UI) related tests.
- Utilized bug tracking tools to log defects, providing detailed descriptions, steps to reproduce, as well as priority and severity levels.
- Gained hands-on experience by testing different types of module from website like <https://the-internet.herokuapp.com/> , and UI Test from <https://bstackdemo.com/> where I performed different tests and practiced both positive and negative testing scenarios.

- Performed end-to-end UI testing for our client, Future Minds Education Consultancy using the Cypress framework. Automated key user flows like navigation, forms, and responsiveness across devices. Worked with developers to identify and fix UI bugs for a seamless user experience.

4.3 Task/Activities Performed

My Internship at Avoloft Technologies included several key areas that were designed to simulate real-world testing scenarios and equip me with practical knowledge. I have got to work on a lot of activities which are listed below:

4.3.1 Manual Testing

Software Testing is a method to check whether the actual software product matches expected requirements and to ensure that software product is defect free. The purpose of software testing is to identify errors, gaps or missing requirements in contrast to actual requirements. I engaged in manual testing of applications which involved creating detailed test plans and test cases, executing these test cases, and identifying and reporting bugs. I performed frontend testing on various web applications, practiced finding locators for UI-related tests, and documented defects comprehensively.

Table 4.1 Test Scenario for Login Functionality

Test Scenario ID	Scenarios	Description	Status
TS01	Successful Login	Verify that the user can log in with valid username and password	Pass
TS02	Invalid Username	Verify that login fails with an incorrect username	Pass
TS03	Invalid Password	Verify that login fails with an incorrect password	Pass
TS04	Empty Username Field	Verify that login is not allowed with an empty username	Pass
TS05	Empty Password Field	Verify that login is not allowed with an empty password	Pass
TS06	Empty Username & Password	Verify that login is not allowed when both fields are empty	Pass
TS07	Password Masking	Verify that the password is masked while typing	Pass
TS08	Forgot Password Link	Verify that the 'Forgot your password?' link redirects to the password reset page	Pass

Table 4.2 Test Case for Login Functionality

Test Case ID	Test Description	Pre-Requisites	Test Steps	Test Data	Expected Outcome	Actual Outcome	Result
TC_001	Login Screen Verification	Valid URL and network connection	Open the application (Orange HRM)	Not Applicable	Username,password field, remember me and login button are visible and clickable	As Expected	Pass
TC_002	Verify login with valid credentials	1. Open the application (Orange HRM) 2. Login Credentials for existing account is required	1. Enter a valid Username in the Username field. 2. Enter a valid password in the password field 3. Click on a login button	1. Username : admin 2. Password : admin123	1. User should get logged in and taken to a (Dashboard)	As Expected	Pass
TC_003	Check login with invalid Username	1. Open the application (Orange HRM)	1. Enter a Invalid Username in the Username field. 2. Enter a valid password in the password field 3. Click on a login button	1. Username : admin12 2. Password : admin123	1. Error message should be displayed with invalid credentials	As Expected	Pass
TC_004	Check login with invalid password	1. Open the application (Orange HRM)	1. Enter a valid Username in the Username field. 2. Enter a invalid password in the password field 3. Click on a login button	1. Username : admin 2. Password : admin@123	1. Error message should be displayed with invalid credentials	As Expected	Pass
TC_005	Check login with invalid credentials	1. Open the application (Orange HRM)	1. Enter a Invalid Username in the Username field. 2. Enter a invalid password in the password field 3. Click on a login button	1. Username : admin123 2. Password : admin123	1. Error message should be displayed with invalid credentials	As Expected	Pass
TC_006	Check login with empty data	1. Open the application (Orange HRM)	1. Don't enter anything in the username field 2. Don't enter anything in the password field 3. Click on a login button		1. Error message should be displayed with Username and Password is required	As Expected	Pass
TC_007	Check login with no Username	1. Open the application (Orange HRM)	1. Don't enter anything in the username field 2. Enter a password 3. Click on a login button	1. Password : admin123	1. Error message should be displayed with username is required	As Expected	Pass
TC_009	Check login with no password	1. Open the application (Orange HRM)	1. Enter a Username in the username field 2. Don't enter anything in the password field 3. Click on a login button	1. Username: admin	1. Error message should be displayed with password is required	As Expected	Pass



Login

 Username

Required

 Password

Required

Login

Figure 4.1 Login Screen of Orange HRM

4.3.2 Bug Reporting

During my internship, I used Cypress for automation testing to ensure the application's stability. While testing various user flows, I identified several bugs, including a critical issue in the Change Password feature. Although the system confirmed a successful update, users were unable to log in with the new password highlighting a mismatch between front-end feedback and backend behavior.

Table 4.3 Bug Reporting of Password Change Functionality

Field	Details
Bug Title	Password Change Success Message Misleading Cannot Log In with New Password
Summary	After changing the password, the system shows success but the user cannot log in with the new password.
Application	OrangeHRM Open Source Demo
URL	https://opensource-demo.orangehrmlive.com/web/index.php/dashboard/index#
Browser	Google Chrome v124.0
Operating System	Windows 10
Testing Tool	Cypress
Preconditions	User is logged in with valid credentials and has access to change the password
Steps to Reproduce	1. Log in with valid credentials 2. Go to 'My Info' > 'Change Password' 3. Change password and submit 4. Log out 5. Attempt login with new password
Expected Result	User should be able to log in with the new password after successful password change



Actual Result	Login fails with the new password; old password still works
Severity	Medium to High
Priority	High
Additional	No error shown during password change; possible backend persistence issue or demo environment limitation

```

URL_Testing.cy.js  LoginTest.cy.js  Dashboard.cy.js  DatePicker.cy.js  PasswordChange.cy.js  cypress.config.js
1  describe('title: "Password Change of HRM"', fn: () : void => {
2    const login = (username) : void => { Show usages
3      cy.session(username, {setup: () : void => {
4        cy.visit('https://opensource-demo.orangehrmlive.com/web/index.php')
5        cy.get(selector: "input[name='username']").type(username);
6        cy.get(selector: "input[name='password']").type(text: 'admin123');
7        cy.get(selector: "button[type='submit']").click();
8      })
9    }
10   beforeEach(fn: () : void => {
11     login(username: "Admin")
12   })
13
14   it('title: "Changing password", config: () : void => {
15     cy.visit('https://opensource-demo.orangehrmlive.com/web/index.php/dashboard/index')
16     cy.get(selector: ".oxd-userdropdown-name").click()
17     cy.contains(selector: "a", text: "Change Password").click()
18     cy.url().should(chainer: "eq", value: "https://opensource-demo.orangehrmlive.com/web/index.php/pim/updatePassword")
19     cy.get(selector: ".orangehrm-card-container > .oxd-text--h6").should(chainer: "have.text", value: "Update Password")
20
21     cy.get(selector: ':nth-child(1) > .oxd-grid-2 > :nth-child(2) > .oxd-input-group > :nth-child(2) > .oxd-input')
22       .type(text: "admin123")
23
24     cy.get(selector: '.user-password-cell > .oxd-input-group > :nth-child(2) > .oxd-input').type(text: "admin@123#")
25
26     cy.get(selector: '.user-password-row > .oxd-grid-2 > :nth-child(2) > .oxd-input-group > :nth-child(2) > .oxd-input')
27       .type(text: "admin@123#")
28
29     cy.get(selector: '.oxd-button--secondary').click()
30
31     cy.get(selector: ".oxd-toast") Chainable<jQuery<...>>
32       .should(chainer: 'be.visible') Chainable<Subject>
33       .and(chainer: "contain.text", value: "Success")
34   })
35 }
36
37 callback for describe() -> callback for it()

```

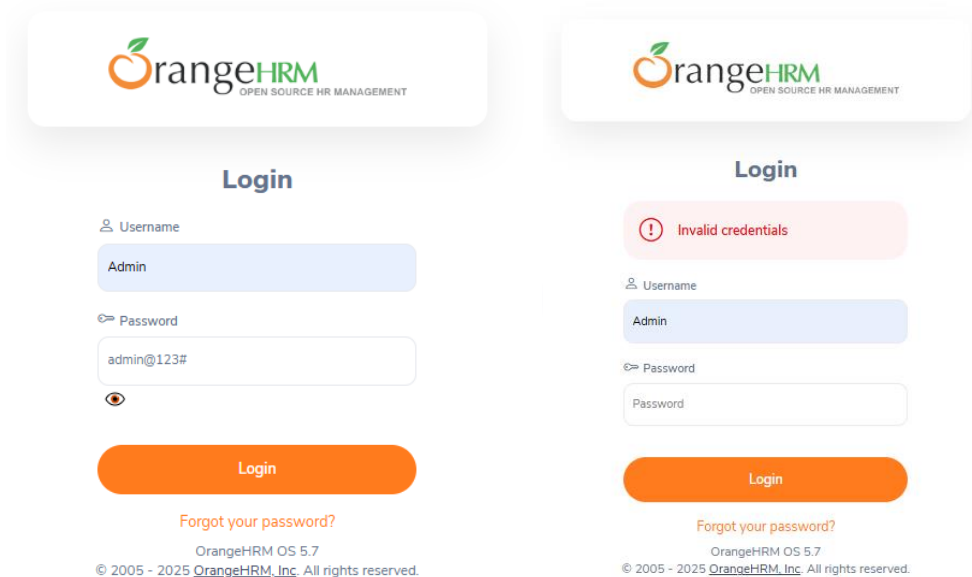


Figure 4.2 Login Dashboard with details of Orange HRM

4.3.3 API Testing

Postman is a popular API development and testing tool that allows you to send HTTP requests, analyze responses, and collaborate with your team. It provides a user-friendly interface for constructing and sending requests to API endpoints, making it easier to

interact with APIs during development.

To work with Postman in JavaScript, we can use the Postman API, which exposes methods and properties to interact with the Postman application programmatically. With the Postman API, you can automate tasks, integrate Postman into your development workflow, and extend its functionality. I learned to perform API testing using Postman by creating and executing various API requests. This experience helped me understand how to validate the functionality, reliability, and performance of APIs.

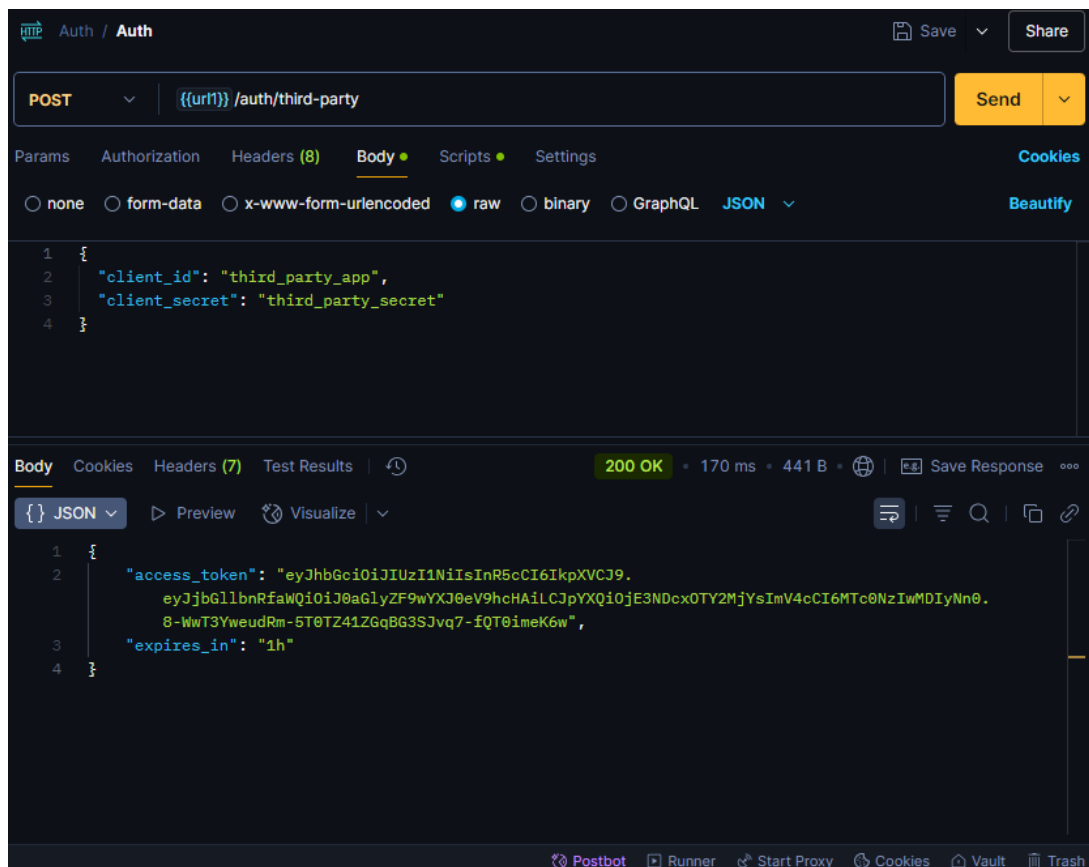


Figure 4.3 OAuth2.0 authorization using client id and client secret key

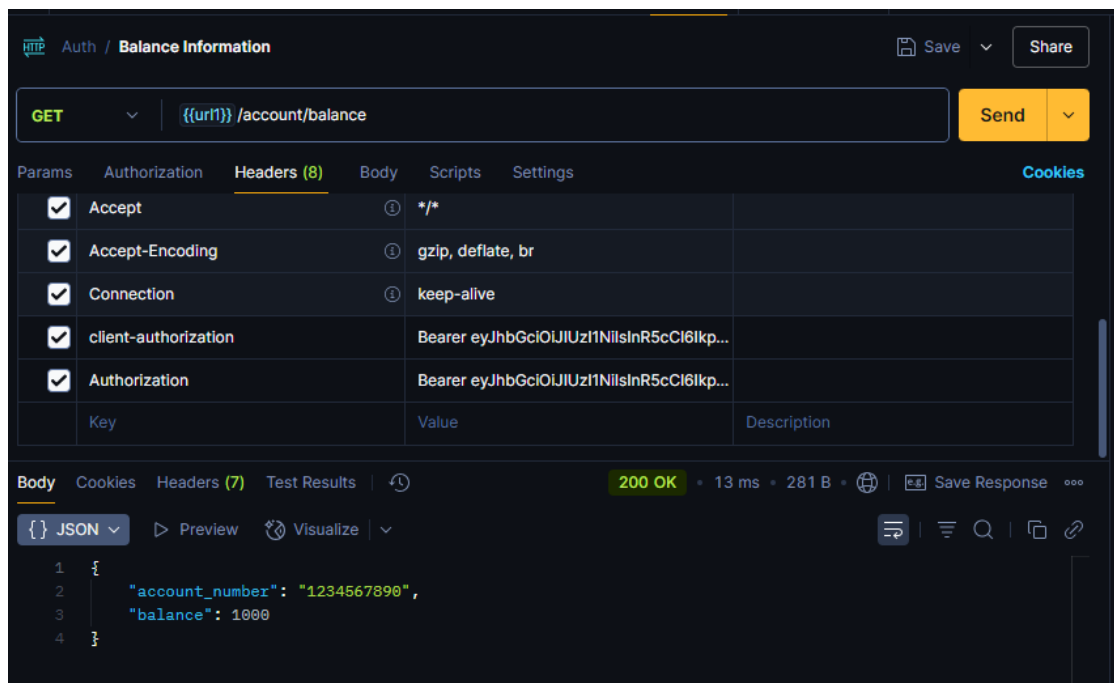


Figure 4.4 Fetched account number and balance of authenticated user

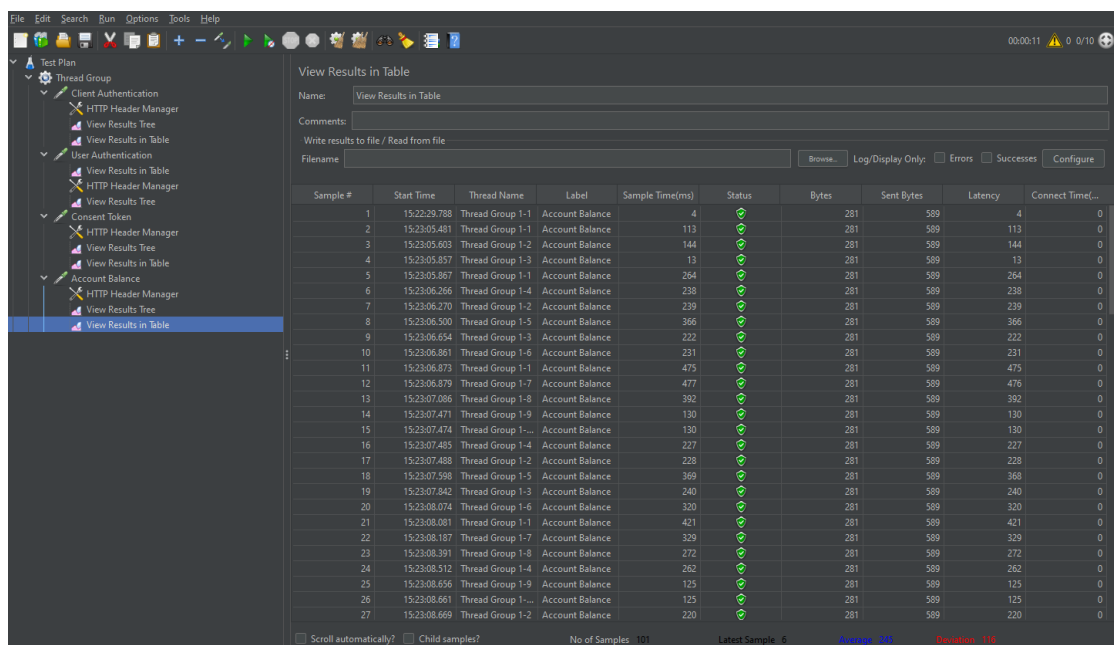


Figure 4.5 Load Testing of Mock-Bank API

4.3.4 Automation Testing

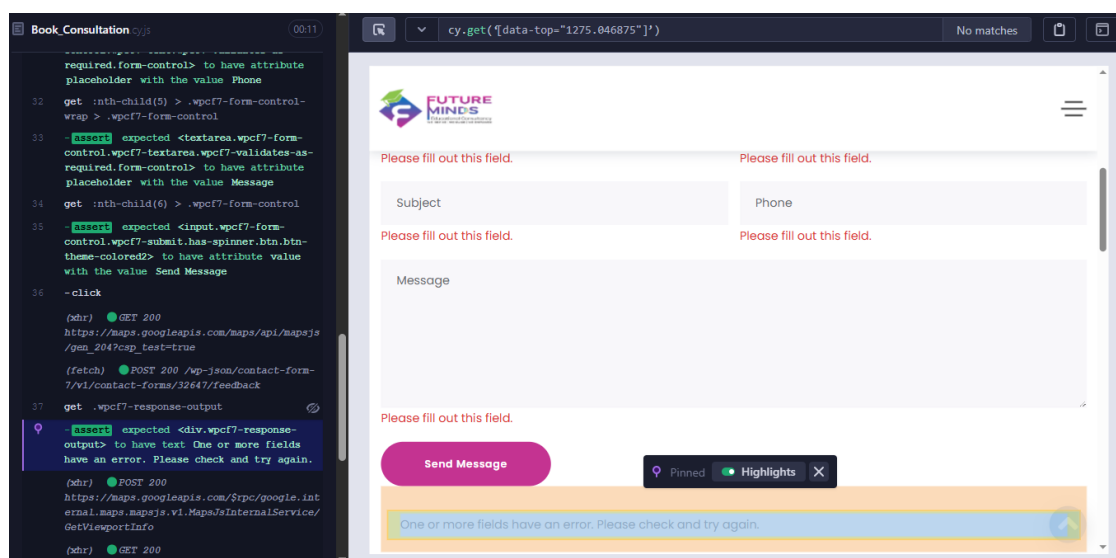
Automation testing is the process of testing software using an automation tool to find defects. In this process, testers execute the test scripts and generate the test results automatically by using automation tools. Some of the famous automation testing tools

for functional testing are QTP/UFT, Cypress, Selenium and Playwright. Automated software testing's main benefit is that it simplifies as much manual effort as possible. If unit testing consumes a large percentage of a QA team's resources, for example, then this process should be evaluated as a candidate for automation. Automated testing tools execute tests, report outcomes and compare results with earlier test runs. Automated tests can run repeatedly at any time of day. This approach fits in with continuous testing and software development practices.

```

1 describe('Session usage example', fn() => {
2   const login = (username) => {
3     cy.session(username, setup() => {
4       cy.visit('https://bstackdemo.com/signin');
5
6       cy.get(selector: '#username').type(username);
7       cy.get(selector: '#username').trigger('keydown', {keyCode: 9});
8       cy.get(selector: '#password').type('testingisfun99\n');
9       cy.get(selector: '#login-btn').click({force: true});
10
11       cy.get(selector: 'span.username', {timeout: 10000}).should('have.text', username);
12     }, {
13       validate() {
14         cy.get(selector: '.username').should('have.text', username);
15       }
16     });
17   };
18
19   it('adds product to cart', config() => {
20     login('demouser')
21     cy.visit('https://bstackdemo.com');
22     cy.get(selector: 'div[data-sku="iPhone12-device-info.png"] div')
23       .filter(selector: ':contains("Add to cart")')
24       .eq(index: 2)
25       .click();
26
27     //this allows to choose the select after the filter is used.
28   });
29 }
  
```

Figure 4.6 Scripts to automate login page using sessions



```

Book_Consultation.cy.js x
1 describe('title: "Book_Consultation"', fn: () : void => {
2   it('title: "Testing with all fields empty"', config: () : void => {
27     cy.get(selector: "div.col-sm-12") Chainable<JQuery<...>>
28       .should(chainer: 'have.length', value: 2) Chainable<Subject>
29       .and(chainer: 'be.visible')
30
31     //NOTE: Testing with a empty data in each input field
32
33     cy.get(selector: ':nth-child(1) > .wpcf7-form-control-wrap > .wpcf7-form-control')
34       .should(chainer: "have.attr", value: "placeholder", match: "Your Name")
35
36     cy.get(selector: ':nth-child(2) > .wpcf7-form-control-wrap > .wpcf7-form-control')
37       .should(chainer: "have.attr", value: "placeholder", match: "Enter Email")
38
39     cy.get(selector: ':nth-child(3) > .wpcf7-form-control-wrap > .wpcf7-form-control')
40       .should(chainer: "have.attr", value: "placeholder", match: "Subject")
41
42     cy.get(selector: ':nth-child(4) > .wpcf7-form-control-wrap > .wpcf7-form-control')
43       .should(chainer: "have.attr", value: "placeholder", match: "Phone")
44
45     cy.get(selector: ':nth-child(5) > .wpcf7-form-control-wrap > .wpcf7-form-control')
46       .should(chainer: "have.attr", value: "placeholder", match: "Message")
47
48     cy.get(selector: ':nth-child(6) > .wpcf7-form-control') Chainable<JQuery<...>>
49       .should(chainer: "have.attr", value: "value", match: "Send Message") Chainable<Subject>
50       .click()
51
52     cy.get(selector: '.wpcf7-response-output')
53       .should(chainer: "have.text", value: "One or more fields have an error. Please check and try again")
54   })
55 })

```

Figure 4.7 UI Testing of Book Consultation service of future minds

4.3.5 Version Control System

Gained hands-on experience with version control using Git, where I managed and tracked code changes across multiple projects. This involved working with Git commands to commit, merge, branch, and resolve conflicts, ensuring smooth collaboration within development teams. I also became proficient in using platforms like GitHub for remote repository management, where I contributed to codebases, reviewed pull requests, and maintained proper versioning. It allowed me to effectively collaborate on projects, maintain code history.

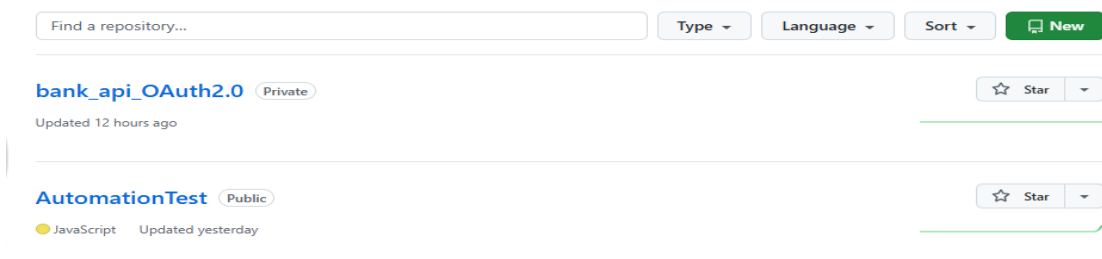


Figure 4.8 Pushed code in to GitHub

CHAPTER FIVE: CONCLUSION AND LESSON LEARNT

5.1 Conclusion

Overall, my internship at Avoloft Technologies as a Quality Assurance Intern has been a transformative experience that bridged the gap between academic knowledge and practical application. In this period of internship, I have tried my best to learn and grab knowledge from my mentor and tried to solve real-world problems.

I gained hands-on experience in manual testing, API testing using Postman, and there is so many more to learn. The structured training modules and practice projects allowed me to develop skills in writing test cases, executing tests, and reporting bugs efficiently. This internship has helped me with the necessary tools and techniques to ensure the quality and reliability of software applications. It has also reinforced the importance of a thorough and systematic approach to testing in delivering high-quality software products. Overall, my time at Avoloft Technologies has been invaluable in preparing me for a future career in software testing and quality assurance.

5.2 Lesson Learnt

Throughout my internship, I learned several valuable lessons that will stay with me as I continue my career in Quality Assurance.

- 1) I also learned various methods of software testing.
- 2) I developed the interpersonal, managerial and entrepreneurial skills that are crucial in every professional and employee's career development.
- 3) I learned the importance of time and punctuality.
- 4) I learned about QA services.
- 5) I learned the importance of being adaptable and open to learning new technologies and frameworks.
- 6) I learned the significance of effective communication and collaboration within a team.
- 7) I learned about the automation testing and test scripts in a postman.
- 8) I learned about the performance and reliability of software testing.

I gained a deep appreciation for the significance of thorough testing and quality assurance in software development.

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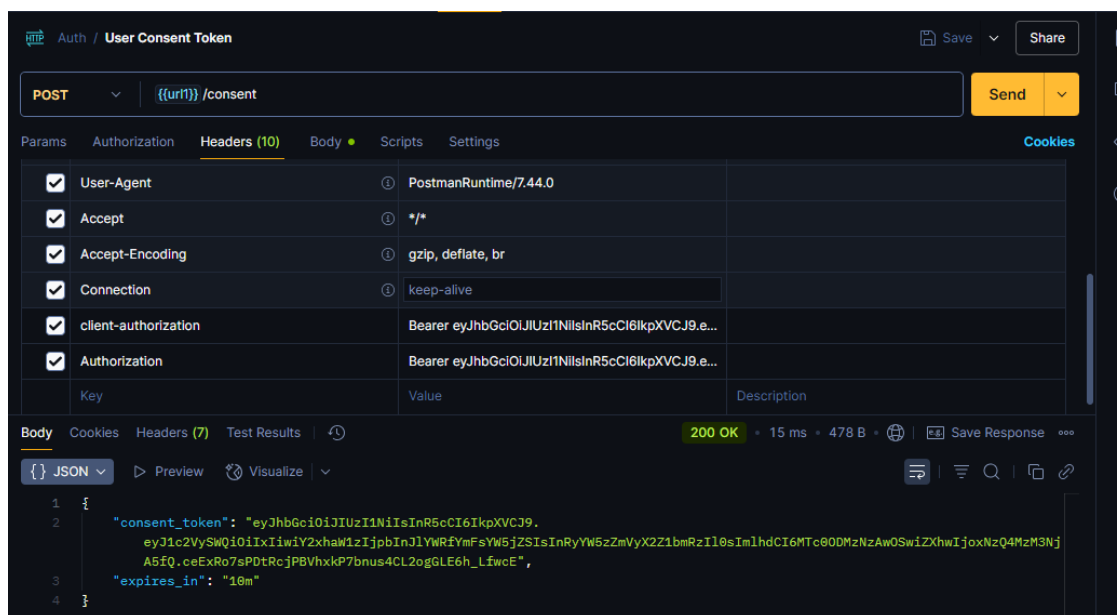
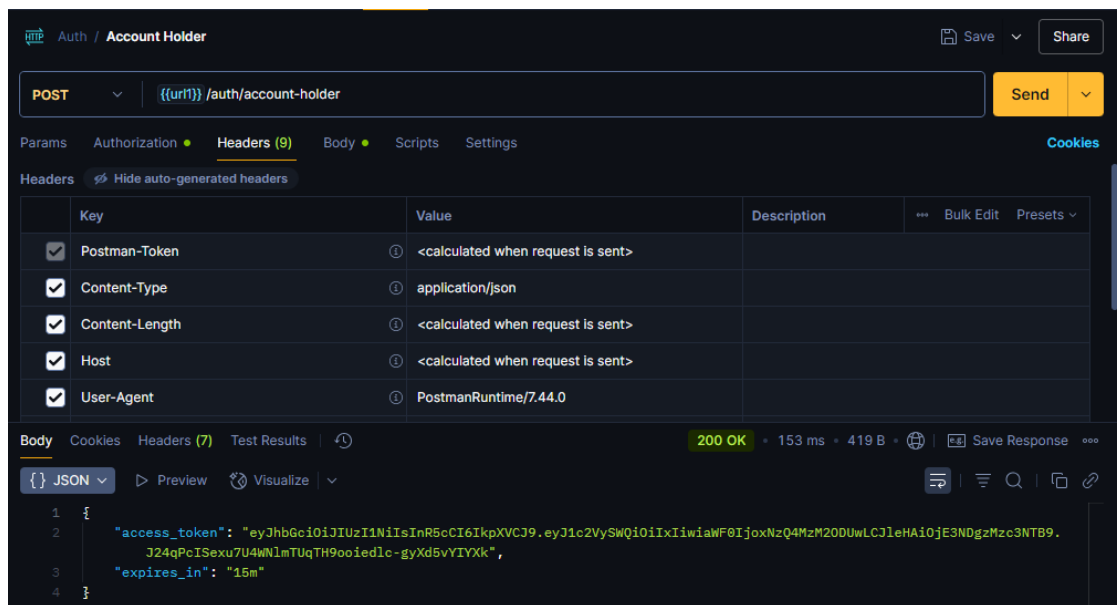
APPENDICES

A. Weekly Log Sheet

Week	Start Date	End Date	Tasks Performed
1	2024-12-17	2024-12-23	1) Introduction to QA and QC, its process review and walk through 2) SDLC and STLC 3) Types of testing 4) Learned about HTTP status codes
2	2024-12-24	2024-12-31	1) Understood about Manual Testing 2) Design test scenarios 3) Designed test case for Login Functionality 4) Design test case for cardless withdrawal 5) Positive and Negative Testing, Severity and Priority Testing 6) Integrated APIs and Its types in Postman.
3	2025-01-01	2025-01-08	1) Learned about Database and its type of testing 2) Node and it's package manager Setup 3) Learned test scripts of postman and SOAP API
4	2025-01-09	2025-01-16	1) Learned about cypress locators and assertions 2) Learned about different methods used in Cypress and chainers
5	2025-01-18	2025-01-25	1) Learned basics of cypress and how to test applications 2) Automated Browser Stack Demo Login automation. 3) Demo of QA automation in Cypress 4) Demo of OpenCart automation
6	2025-01-26	2025-02-02	1) Learned about handling sessions 2) Written scripts for OrangeHRM Login and

			integrated different functionality like verifying actual user gets logged in, verification of the URL and so on.
7	2025-02-03	2025-02-10	1) Automated upload and download files, for internet-heroku-app and used Mouse Hover, Drag and Drop, Downloading Files, Renaming Files, Right Click and Double Click
8	2025-02-11	2025-02-18	1) Automated add to cart page of Browser Stack demo and securely download files by passing auth parameters in URL along with Dynamic Content that changed in each reload using cypress
9	2025-02-19	2025-02-25	1) Learned about the JMeter for performance testing and stress testing of the application and modules.
10	2025-02-26	2025-03-04	1) Learned about Git basics, Pushed all the projects to the git 2) Cloned repository, Created new git branch 3) Added screenshot of automation code
11	2025-03-05	2025-03-11	1) Pulled request, Code and Branch review by mentor Pulled changes 2) Added files to .gitignore, and pull changes
12	2025-03-12	2025-03-16	1) Created business requirement specification or description document and test plan document along with using basics of JIRA for a project management

B. Screenshots



Requests / List Users

GET {{url}} /users?page=2

Send

Params • Authorization Headers (6) Body Scripts • Settings Cookies

Pre-request

Post-response •

```
1 pm.test("Response Time to be less than 10ms", function () {
2   pm.expect(pm.response.responseTime).to.be.below(100);
3 });
4 pm.test("Test to be 201 Created", () => {
5   pm.expect(pm.response.to.have.status(200))
6 })
7 pm.test("Content-Type", () => {
8   pm.expect(pm.response.to.have.header("Content-Type"))
9 })
10 pm.test("Content-Type header is application/json", () => {
```

Body Cookies Headers (23) Test Results (6/6) 200 OK • 22 ms • 1.62 KB • Save Response

Filter Results

- PASSED Response Time to be less than 10ms
- PASSED Test to be 201 Created
- PASSED Content-Type
- PASSED Content-Type header is application/json
- PASSED Response code to equal 200
- PASSED Body contains some string

load.cy.js ForgotPassword.cy.js Hovers.cy.js logintest.cy.js DynamicContent.cy.js

```
1 describe('Dynamic_Content', fn: () => {
2   it('Content', config: () => {
3     cy.visit('https://the-internet.herokuapp.com/dynamic_content')
4     cy.get(selector: '.large-2.columns img').should(chainer: 'exist')
5       .and(chainer: 'be.visible')
6     cy.get(selector: '.large-10.columns').should(chainer: 'exist')
7       .and(chainer: 'not.be.empty')
8   })
9
10  it.only('Should change content on refresh', config: () => {
11    cy.visit('https://the-internet.herokuapp.com/dynamic_content')
12    cy.get(selector: '.large-10.columns').invoke(functionName: 'text')
13      .then(fn: (text1 : ReturnType<...> ) => {
14        cy.reload()
15        cy.get(selector: '.large-10.columns').Chainable<JQuery<...>>
16          .invoke(functionName: 'text') Chainable<ReturnType<...>>
17          .should(chainer: (text2 : ReturnType<...> ) => {
18            expect(text2).not.to.eq(text1);
19          });
20      });
21  })
22
23  it('Should change the image on refresh', config: () => {
24    cy.visit('https://the-internet.herokuapp.com/dynamic_content');
25    cy.get(selector: '.large-2.columns img') Chainable<JQuery<...>>
26      .invoke(functionName: 'attr', args: 'src') Chainable<ReturnType<...>>
27      .then(fn: (srcBefore : ReturnType<...> ) => {
28        cy.reload();
29      });
30  })
31})
```

callback for describe() > callback for it() > callback for cy.get(".large-2.columns img")...).then()


```

basicauth.cy.js  contextmenu.cy.js  Dissapering_Elements.cy.js  Drag-n-Drop_FileUpload.cy.js  DragandDrop.cy.js
describe('title: "Drag-n-Drop_FileUpload"', fn: () => {  Nibesh Upadhyay
  it('title: "File Upload by Dragging"', config: () => {
    cy.visit('url: "https://the-internet.herokuapp.com/upload"')
    cy.get(selector: "#drag-drop-upload")
      .attachFile(fixture: 'menu.pdf', processingOpts: {subjectType: "drag-n-drop"})
    cy.get(selector: '#drag-drop-upload > .dz-preview > .dz-details > .dz-filename')
      .should(chainer: 'have.text', value: 'menu.pdf')
  })

  // Checking for renaming a file in it.....
  it.only('title: "Renaming a file in it"', config: () => {
    cy.visit('url: "https://the-internet.herokuapp.com/upload"')
    cy.get(selector: "#drag-drop-upload")
      .attachFile(fixture: {filePath: 'menu.pdf', fileName: 'menu1.pdf'}, processingOpts: {subjectType: 'drag-n-drop'})

    cy.get(selector: '#drag-drop-upload > .dz-preview > .dz-details > .dz-filename > span')
      .should(chainer: 'have.text', value: 'menu1.pdf')
  })
})

```

Update Password

Username

Admin

Current Password*

admin123

Password*

admin@123#

Strong

Confirm Password*

admin@123#

For a strong password, please use a hard to guess combination of text with upper and lower case characters, symbols and numbers

* Required

Cancel

Save

```

36
37     cy.get(selector: ".oxd-userdropdown-name").click()
38     cy.contains(selector: "a", text: "Logout").click()
39
40     cy.url().should(chainer: 'eq', value: "https://opensource-demo.orangehrmlive.com/web/index.php/auth/login")
41     cy.get(selector: "input[name='username']").type(text: "Admin");
42     cy.get(selector: "input[name='password']").type(New_Password);
43     cy.get(selector: "button[type='submit']").click();
44
45     cy.url().should(chainer: 'eq', value: "https://opensource-demo.orangehrmlive.com/web/index.php/dashboard/index")
46
47   })
48 }

```

```
cy.get(selector: ':nth-child(1) > .oxd-grid-2 > :nth-child(2) > .oxd-input-group > :nth-child(2) > .oxd-input')
  .type(text: "admin123")
cy.get(selector: '.user-password-cell > .oxd-input-group > :nth-child(2) > .oxd-input').type(text: "admin123")

cy.get(selector: '.user-password-row > .oxd-grid-2 > :nth-child(2) > .oxd-input-group > :nth-child(2) > .oxd-input')
  .type(text: "admin123")

cy.get(selector: '.oxd-button--secondary').click()

cy.get(selector: ".oxd-toast") Chainable<JQuery<...>>
  .should(chainer: "be.visible") Chainable<Subject>
  .and(chainer: "contain.text", value: "Error")

})
})
```